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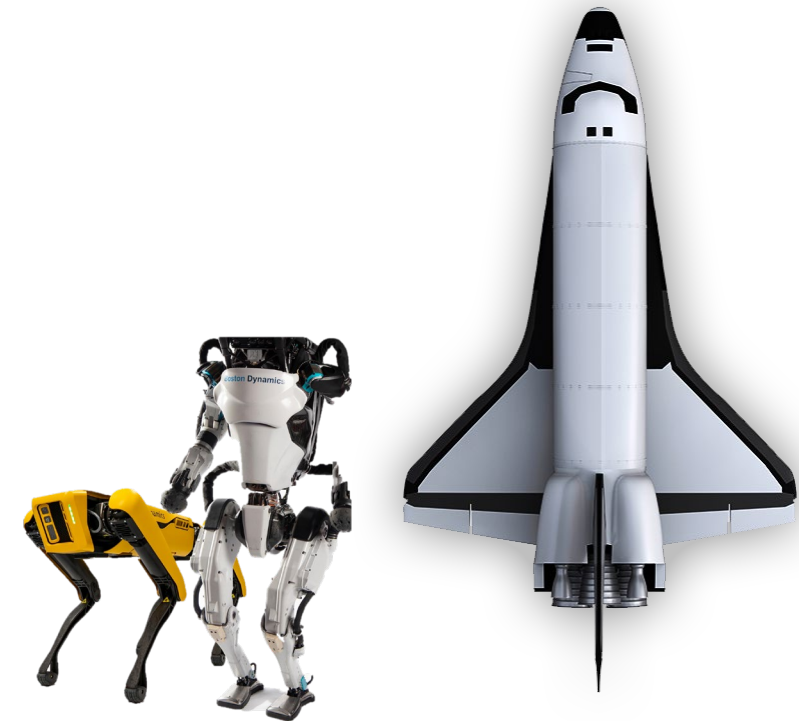
ADVANCED CONTROL SYSTEM DESIGN

Lecture 5

Controllability in State- Space



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INTRODUCTION

The concepts of **controllability** and **observability** were introduced by **Rudolf Emil Kalman**. They play an important role in the design of control systems in state space.

IMPORTANCE

The conditions of controllability and observability may govern the existence of a **complete solution** to the control system design problem. The solution to this problem may not exist if the system considered is not controllable. An uncontrollable system has a subsystem that is physically **disconnected** from the input.

CONCEPT

A system is said to be **controllable** at time t_0 if it is possible by means of an unconstrained control vector to transfer the system from any **initial state** $x(t_0)$ to any **other state** in a finite interval of time.

Controllability

SYSTEMS MODELS

Although most physical systems are controllable and observable, corresponding mathematical models may not possess the property of controllability and observability.

TYPES

Controllability can be divided into state controllability and output controllability. These two types can be found using several approaches. The state controllability affects the stabilizability of the system.

State Controllability

CONCEPT

A system is said to be **state controllable** at time $t = t_0$ if it is possible by means of an unconstrained control vector to transfer the system from any initial state $x(t_0)$ to any other state in a finite interval of time $t_0 \leq t \leq t_1$. If every state is controllable, then the system is said to be completely state controllable.

CONDITION

The system represented by the state space equations is completely state controllable if and only if the vectors $\mathbf{B}$, $\mathbf{AB}$, ..., $\mathbf{A}^{n-1}\mathbf{B}$ are linearly independent, or the $n \times n$ matrix, $\mathbf{M}$, is a full rank.

$$\mathbf{M} = \begin{bmatrix} \mathbf{B} & \mathbf{AB} & \dots & \mathbf{A}^{n-1}\mathbf{B} \end{bmatrix}$$

State Controllability

EXAMPLE 1

Determine the state controllability of the systems below:

$$1. \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u$$

$$2. \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

$$3. \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 2 \\ 2 & -1 & -4 \\ -1 & -2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 5 \end{bmatrix} u$$

$$1. |\mathbf{M}| = |\mathbf{B} \quad \mathbf{AB}| = \begin{vmatrix} 1 & 1 \\ 0 & 0 \end{vmatrix} = 0 \Rightarrow \text{Not completely state controllable.}$$

$$2. |\mathbf{M}| = |\mathbf{B} \quad \mathbf{AB}| = \begin{vmatrix} 0 & 1 \\ 1 & -1 \end{vmatrix} = -1 \Rightarrow \text{Completely state controllable.}$$

State Controllability

3. $|\mathbf{M}| = |\mathbf{B} \quad \mathbf{AB} \quad \mathbf{A}^2\mathbf{B}| = \begin{vmatrix} 0 & 11 & -4 \\ 1 & -21 & 31 \\ 5 & 3 & 34 \end{vmatrix} = 899 \Rightarrow$ Completely state controllable.

EXERCISE 1

Determine the state controllability of the systems below:

$$1. \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 2 \\ 5 \end{bmatrix} u$$

$$2. \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 2 \\ 0 \end{bmatrix} u$$

$$3. \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 4 \\ 3 \end{bmatrix} u$$

Alternative Forms of State Controllability

CONCEPT

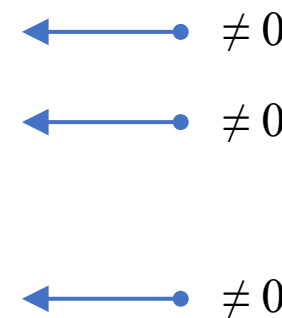
This approach relies on investigating the state controllability of systems that are either in the **diagonal** form or the **Jordan** form.

DIAGONAL CONDITION

The condition of complete state controllability is that if the eigenvectors of **A** are distinct, and the matrix **D** = **P**⁻¹**A****P** is in the **diagonal** form, then the system is completely state controllable if and only if **no row** of **E** = **P**⁻¹**B** has all **zero** elements, where:

$$\dot{\mathbf{z}} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}\mathbf{z} + \mathbf{P}^{-1}\mathbf{B}u = \mathbf{D}\mathbf{z} + \mathbf{E}u$$

$$\begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \\ \vdots \\ \dot{z}_n \end{bmatrix} = \begin{bmatrix} -\lambda_1 & 0 & 0 & 0 \\ 0 & -\lambda_2 & 0 & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & -\lambda_n \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{bmatrix} + \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{bmatrix} u$$



Alternative Forms of State Controllability

JORDAN CONDITIONS

The condition of complete state controllability is that if the eigenvectors of $\mathbf{A}$ are repeated, and the matrix $\mathbf{J} = \mathbf{S}^{-1}\mathbf{A}\mathbf{S}$ is in the **Jordan** form and $\mathbf{G} = \mathbf{S}^{-1}\mathbf{B}$, then the system is completely state controllable if and only if:

1. No two **Jordan blocks** in $\mathbf{J}$ are associated with the same eigenvalues.
2. The elements of any row of $\mathbf{G}$ that correspond to the last row of each **Jordan block** are not all zero.
3. The elements of each row of $\mathbf{G}$ that correspond to **distinct eigenvalues** are not all zero.

$$\dot{\mathbf{z}} = \mathbf{S}^{-1}\mathbf{A}\mathbf{S}\mathbf{z} + \mathbf{S}^{-1}\mathbf{B}u = \mathbf{J}\mathbf{z} + \mathbf{G}u$$

$$\begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \\ \dot{z}_3 \\ \dot{z}_4 \\ \dot{z}_5 \\ \dot{z}_6 \\ \dot{z}_n \end{bmatrix} = \begin{bmatrix} -\lambda_1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\lambda_1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\lambda_1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\lambda_4 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\lambda_4 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\lambda_5 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\lambda_n \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \\ z_4 \\ z_5 \\ z_6 \\ z_n \end{bmatrix} + \begin{bmatrix} g_1 \\ g_2 \\ g_3 \\ g_4 \\ g_5 \\ g_6 \\ g_n \end{bmatrix} u$$

$\leftarrow \bullet \neq 0$
 $\leftarrow \bullet \neq 0$
 $\leftarrow \bullet \neq 0$
 $\leftarrow \bullet \neq 0$

Alternative Forms of State Controllability

EXAMPLE 2

Determine the state controllability of the systems below using the alternative forms:

$$1. \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 2 \\ 5 \end{bmatrix} u$$

By inspection: Completely state controllable.

$$2. \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 2 \\ 0 \end{bmatrix} u$$

By inspection: Not completely state controllable.

$$3. \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 4 \\ 3 \end{bmatrix} u$$

By inspection: Completely state controllable.

Alternative Forms of State Controllability

$$4. \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 4 & 2 \\ 0 & 0 \\ 3 & 0 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

By inspection: Not completely state controllable.

$$5. \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \\ \dot{x}_5 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 0 & 0 & 0 \\ 0 & -2 & 1 & 0 & 0 \\ 0 & 0 & -2 & 0 & 0 \\ 0 & 0 & 0 & -5 & 1 \\ 0 & 0 & 0 & 0 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 0 & 0 \\ 3 & 0 \\ 0 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

By inspection: Completely state controllable.

$$6. \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \\ \dot{x}_5 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 0 & 0 & 0 \\ 0 & -2 & 1 & 0 & 0 \\ 0 & 0 & -2 & 0 & 0 \\ 0 & 0 & 0 & -5 & 1 \\ 0 & 0 & 0 & 0 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} + \begin{bmatrix} 4 \\ 2 \\ 1 \\ 3 \\ 0 \end{bmatrix} u$$

By inspection: Not completely state controllable.

Alternative Forms of State Controllability

$$7. \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -24 & -26 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix} u$$

Inspection cannot be used. Perform diagonalization.

$$|\lambda \mathbf{I} - \mathbf{A}| = \begin{vmatrix} \lambda & -1 & 0 \\ 0 & \lambda & -1 \\ 24 & 26 & \lambda + 9 \end{vmatrix} = \lambda^3 + 9\lambda^2 + 26\lambda + 24$$

$$\lambda_1 = -2, \lambda_2 = -3, \lambda_3 = -4$$

Since the eigenvalues are real and distinct, then assume that:

$$\mathbf{x} = \mathbf{P}\mathbf{z}$$

$$\mathbf{P} = \begin{bmatrix} 1 & 1 & 1 \\ \lambda_1 & \lambda_2 & \lambda_3 \\ \lambda_1^2 & \lambda_2^2 & \lambda_3^2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ -2 & -3 & -4 \\ 4 & 9 & 16 \end{bmatrix}$$

$$\dot{\mathbf{z}} = \mathbf{P}^{-1} \mathbf{A} \mathbf{P} \mathbf{z} + \mathbf{P}^{-1} \mathbf{B} u$$

$$\begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \\ \dot{z}_3 \end{bmatrix} = \begin{bmatrix} 6 & 3.5 & 0.5 \\ -8 & -6 & -1 \\ 3 & 2.5 & 0.5 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -24 & -26 & -9 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ -2 & -3 & -4 \\ 4 & 9 & 16 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} + \begin{bmatrix} 6 & 3.5 & 0.5 \\ -8 & -6 & -1 \\ 3 & 2.5 & 0.5 \end{bmatrix} \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix} u$$

$$\begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \\ \dot{z}_3 \end{bmatrix} = \begin{bmatrix} -2 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -4 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} + \begin{bmatrix} 10.5 \\ -18 \\ 7.5 \end{bmatrix} u$$

By inspection: Completely state controllable.

Alternative Forms of State Controllability

$$8. \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -3 & -7 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} u$$

Inspection cannot be used. Perform diagonalization.

$$|\lambda \mathbf{I} - \mathbf{A}| = \begin{vmatrix} \lambda & -1 & 0 \\ 0 & \lambda & -1 \\ 3 & 7 & \lambda + 5 \end{vmatrix} = \lambda^3 + 5\lambda^2 + 7\lambda + 3$$

$$\lambda_1 = \lambda_2 = -1, \lambda_3 = -3$$

Since the eigenvalues are real and repeated, then assume that:

$$\mathbf{x} = \mathbf{S}\mathbf{z}$$

$$\mathbf{S} = \begin{bmatrix} 1 & 0 & 1 \\ \lambda_1 & 1 & \lambda_3 \\ \lambda_1^2 & 2\lambda_1 & \lambda_3^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & -3 \\ 1 & -2 & 9 \end{bmatrix}$$

$$\dot{\mathbf{z}} = \mathbf{S}^{-1} \mathbf{A} \mathbf{S} \mathbf{z} + \mathbf{S}^{-1} \mathbf{B} u$$

$$\begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \\ \dot{z}_3 \end{bmatrix} = \begin{bmatrix} 0.75 & -0.5 & -0.25 \\ 1.5 & 2 & 0.5 \\ 0.25 & 0.5 & 0.25 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -3 & -7 & -5 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & -3 \\ 1 & -2 & 9 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} + \begin{bmatrix} 0.75 & -0.5 & -0.25 \\ 1.5 & 2 & 0.5 \\ 0.25 & 0.5 & 0.25 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} u$$

$$\begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \\ \dot{z}_3 \end{bmatrix} = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -3 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} + \begin{bmatrix} 1.5 \\ 3 \\ 0.5 \end{bmatrix} u$$

By inspection: Completely state controllable.

State Controllability in the s-Plane

CONDITION

A necessary and sufficient condition for complete state controllability is that **no cancellation** occur in the transfer function or transfer matrix. If cancellation occurs, the system cannot be controlled in the direction of the canceled mode.

State Controllability in the s-Plane

EXAMPLE 3

Determine the state controllability of the following system using multiple approaches:

$$\frac{Y(s)}{U(s)} = \frac{s + 2.5}{(s + 2.5)(s - 1)}$$

Due to the pole-zero cancellation, there is a loss of the control of these states. Thus, the system is not completely state controllable.

$$\frac{Y(s)}{U(s)} = \frac{\cancel{s + 2.5}}{\cancel{(s + 2.5)}(s - 1)} = \frac{1}{s - 1}$$

The state-space representation of the system (using the n^{th} order approach):

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 2.5 & -1.5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u$$

$$|\mathbf{M}| = |\mathbf{B} \quad \mathbf{AB}| = \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} = 0$$

This confirms that the system is not completely state controllable.

Output Controllability

OUTPUT VS. STATES

In the practical design of a control system, we may want to control the **output** rather than the **state** of the system. Complete state controllability is neither necessary nor sufficient for controlling the output of the system. For this reason, it is desirable to define separately complete output controllability.

CONDITION

A system described in a state space representation is completely output controllable if and only if the **N** matrix with dimension of $m \times (n+1)r$ is of rank m , where the state vector is an n -vector, the control vector is an r -vector, and the output vector is an m -vector.

$$\mathbf{N} = \begin{bmatrix} \mathbf{CB} & \mathbf{CAB} & \dots & \mathbf{CA}^{n-1}\mathbf{B} & \mathbf{D} \end{bmatrix}$$

Output Controllability

EXAMPLE 4

Determine the output controllability of the following system:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 3 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\mathbf{N} = [\mathbf{CB} \quad \mathbf{CAB} \quad \mathbf{CA}^2\mathbf{B} \quad \mathbf{D}]$$

$$\mathbf{N} = \begin{bmatrix} 0 & 1 & 0 & 2 & 0 & 4 & 0 & 0 \\ 1 & 0 & 2 & 0 & 4 & 0 & 0 & 0 \end{bmatrix} \Rightarrow \text{The rank is } 2 = m. \text{ Thus, the system is completely output controllable.}$$

Stabilizability

CONDITIONS

If the system is **fully state controllable**, then the system is stabilizable. For a **partially state controllable** system, if the uncontrollable modes are stable, and the unstable modes are controllable, then the system is stabilizable.

MODES

The stabilizability can be investigated by testing the controllability of the modes that correspond to the eigenvalues $(\lambda_1, \lambda_2, \dots, \lambda_n)$. If the matrix $\mathbf{T}_i$ has rank of n , then the corresponding mode is controllable.

$$\mathbf{T}_i = [\lambda_i \mathbf{I} - \mathbf{A} \quad \mathbf{B}]$$

EXAMPLE 5

Investigate the stabilizability of the following systems:

$$1. \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 2 \\ 5 \end{bmatrix} u$$

$$2. \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u$$

First system:

$$\mathbf{M} = [\mathbf{B} \quad \mathbf{AB}] = \begin{bmatrix} 2 & -2 \\ 5 & -10 \end{bmatrix}$$

The rank of $\mathbf{M}$ is 2. Thus, the system is fully-state controllable and stabilizable.

Second system:

$$\mathbf{M} = [\mathbf{B} \quad \mathbf{AB}] = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

The rank of $\mathbf{M}$ is 1. Thus, the system is not fully-state controllable.

Stabilizability

$$|\lambda \mathbf{I} - \mathbf{A}| = \left| \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \right| = \begin{vmatrix} \lambda - 1 & 0 \\ 0 & \lambda + 1 \end{vmatrix} = 0$$

$$\lambda_1 = 1, \lambda_2 = -1$$

$$\mathbf{T}_1 = [\lambda_1 \mathbf{I} - \mathbf{A} \quad \mathbf{B}] = \left[\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right] = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix}$$

$\mathbf{T}_1$ is a full rank matrix. Thus, the mode unstable, but controllable.

$$\mathbf{T}_2 = [\lambda_2 \mathbf{I} - \mathbf{A} \quad \mathbf{B}] = \left[\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right] = \begin{bmatrix} -2 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$\mathbf{T}_2$ is not a full rank matrix. Thus, the mode is uncontrollable, but it is stable.

Based on the above, the system is stabilizable.

Observability in State-Space

Next Week ...

**DO YOU
HAVE ANY
QUESTIONS**